

UNION

Combines results of two or more SELECT statements and **removes duplicate rows**.

UNION ALL

Combines results of two or more SELECT statements **without removing duplicates**.

INTERSECT

Returns only the rows that exist in **both** result sets.

EXCEPT

Returns rows from the first query that are **not present** in the second query.

Operator	Removes Duplicates	Returns
UNION	Yes	All unique rows
UNION ALL	No	All rows including duplicates
INTERSECT	Yes	Common rows
EXCEPT	Yes	Rows in first query only

```
(SELECT e.empno,e.ename,e.deptno,d.dname
FROM emps e LEFT OUTER JOIN depts d
ON e.deptno = d.deptno)
UNION
(SELECT e.empno,e.ename,e.deptno,d.dname
FROM emps e RIGHT OUTER JOIN depts d
ON e.deptno = d.deptno);
```

```
(SELECT ename FROM emps)
```

UNION

(SELECT dname FROM depts);

(SELECT ename FROM emps)

UNION

(SELECT deptno FROM depts);

(SELECT empno,ename FROM emps)

UNION

(SELECT deptno FROM depts);

-- ERROR : The used SELECT statements have a different number of columns

deptno	SUM(Sal)
20	10875.00
30	9400.00
10	8750.00

SELECT SUM(Sal) FROM emp;

SELECT deptno,SUM(Sal) FROM emp
GROUP BY deptno;

deptno	sum
10	8750
20	10875
30	9400
NULL	29025

SUM(Sal)
29025.00

(SELECT deptno,SUM(Sal) FROM emp
GROUP BY deptno)

UNION

(SELECT NULL,SUM(Sal) FROM emp);

SELECT deptno FROM emps; → 10, 20, 50

SELECT deptno FROM depts; → 10, 20, 30, 40

→ (SELECT deptno FROM emps)

INTERSECT

→ (SELECT deptno FROM depts);

10, 20

(SELECT deptno FROM emps) → 10 20 50
EXCEPT
(SELECT deptno FROM depts) → 10 20 30 40

(SELECT deptno FROM depts) → 10 20 30 40
EXCEPT
(SELECT deptno FROM emps) → 10 20 50